Working with resources

eHealth Master SS2023
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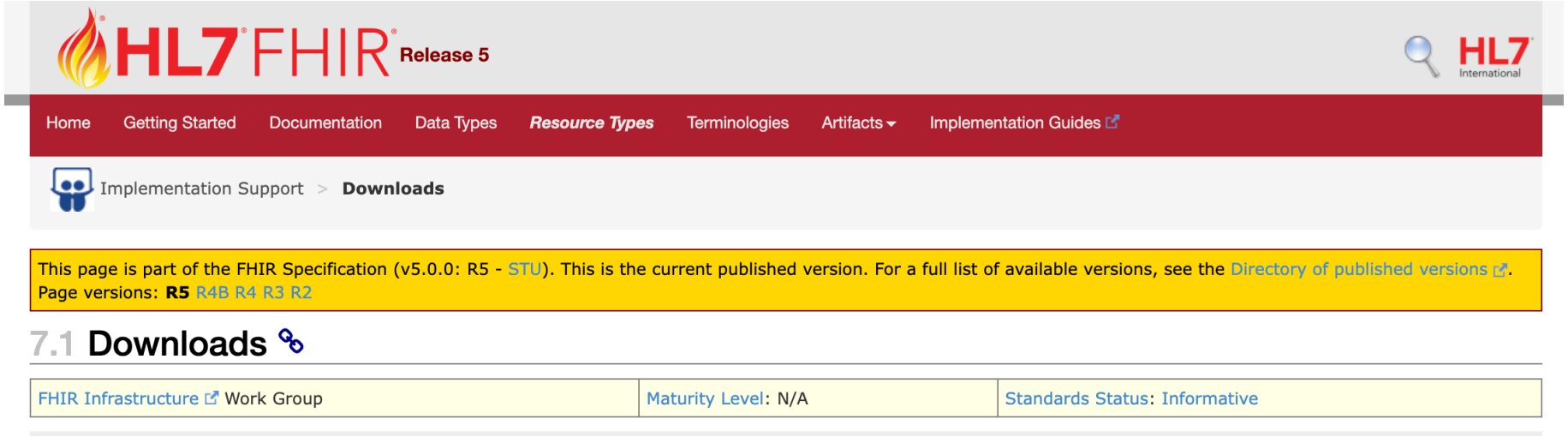
CREATE PATIENT RESOURCE

Prerequsits

- Installed VSCode
- Installed Insomnia or Postman

Download JSON schema

- HL7.org/FHIR
- Documentation / Downloads → Download JSON schemas



Patient Resource

- Make a new folder for your patient resource
- Patient resource (everything beside the Name can be faked):
  - *Name (Vor- und Familienname)*
  - *Geschlecht*
  - *Geburtsdatum*
  - *Telefon*
  - *Email*
  - *Vollständige Adresse inkl. Postleitzahl*

Patient Resource

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- Open folder with VSCode
- Copy fhir.schema.json in folder
  - The file contains ALL FHIR resources

Setting up VSCode to recognize FHIR schema

- Open Command Palette
  - Mac OS: command+shift+P

- Open → Preferences: Open Workspace Settings (JSON)
- This is a JSON by itself with many auto complete functions
- Autocomplete json.schema
- Configure it for FHIR files. Should look like this:
- Create a new file patient.fhir.json

Create patient resource

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- Go to the FHIR patient resource (HL7.org) and make yourself familiar with the patient resource
- Create the patient resource making use of the autocomplete functions and the error messages in case the resource is inconsistent

Create an observation resource

- Now create in the same way a simple observation resource which is validated through the process we followed before
- Status should be ”final”

REST OPERATIONS

Public available FHIR server

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- Https://wiki.hl7.org/Publicly_Available_FHIR_Servers_for_testing

- Institute eHealth FHIR Server:
  - http://diam.fh-joanneum.at:8021/hapi-fhir-jpaserver/
  - http://diam.fh-joanneum.at:8020/hapi-fhir-jpaserver/

POST your FHIR resource

- POST your own created FHIR resource to the Patient endpoint of the following FHIR server:
  - http://diam.fh-joanneum.at:8021/hapi-fhir-jpaserver/
- Check if you get the ”201 Created” status back
- Also check if you got an ”id” for your Patient resource

GET your Patient resource back

- Get your Patient resource back using the “id” of you patient resource
- Check that you receive the status “200 OK”

POST your Observation resource

- Post your Observation resource to the observation endpoint
- Make sure that the reference is linking to your patient resource
- You should again get back the “201 created” message

The Observation was wrong

- Imagine the Observation you just posted had accidently a wrong value of an measurement
- Correct your resource by:
  - Change the status of the observation from “final” to “corrected”
  - Change a value which needs to be corrected
  - Update the resource on the server with the REST PUT command
  - Don’t forget to put the ”id” from the original Observation resource on the server in the update resource in the JSON file as well as in the REST request
- Check if you get the “200 OK” message back
- Check if the version has changed
- Check with _history the full history of the resource

*btw PUT you can also use to create a resource if you have the ”id” already

BUNDLE TUTORIAL

Inform yourself about FHIR Bundles

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https://fhir-drills.github.io/bundle.html

BUNDLES AND TRANSACTIONS

Creating transaction bundle

- Transactions bundles are used to transfer several resource actions to a server
- The server will try to perform all actions (creating resources, updating resources etc.)
- ONLY when all actions of the bundle are successfully performed the transaction is successful (all or nothing)
- In comparison to an collection bundle where the bundle is stored as such

Getting started

- resourceType = Bundle
- Type=Transaction
- Entry 1:
  - Resource → Patient Resource
- Under the resource comes the ”request” which tells the server what to do with the resource
  - ”method”: “POST”
  - “url”: “Patient”

- “fullUrl”: “urn:uuid:patient” needs to be added

Adding another resource

- Entry 2:
  - Resource → Observation
  - Subject
    - “type”: “Patient”
    - “reference”: “urn:uuid:patient”
- Under the resource comes the ”request” which tells the server what to do with the resource (simple POST is enough)

Post the Bundle

- Post the Bundle again to the server
- Check in the reply if the patient ID has been generated
- Get the observation you just posted back from the server and check if the reference to the patient has been correctly made

….it goes on

- Change the observation in your editor and add an value
- Now we want to update this observation again on the server BUT do not want to create a new patient.
- We can change our Patient resource in the Bundle the following way
  - Instead of “method”: “POST”
  - “method”: “PUT”
  - “url”: “Patient?name=IamtheTestUser”
- What does it do? → instead of creating a new patient it checks if the patients exists and updates the observation otherwise creates the patient
- Check what you get back!

Exercise: Another CRUD FHIR tutorial

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- Https://fhir-drills.github.io/simple-patient.html

- And:

https://developer-wp-uks.azurewebsites.net/learn/playing-with-fhir-tut002-creating-and-editing-a-patient/